ESE 1205 One-Shot Notes

Thermodynamics for Energy Engineering · KUET · past papers 2017, 2018, 2019, 2020, 2021, 2022, 2023 · answers from the course slides (Tahir Mahmud, Afsar, Sohag) and the Cengel and Moran-Shapiro textbooks they follow

One-shot notes: ESE 1205 Thermodynamics

Basic words (A1)

System / surroundings / boundary. Closed (control mass: energy only), open (control volume: mass + energy), isolated (nothing). Intensive (T, p, v) vs extensive (m, V, U); specific = extensive/m.

Equilibrium = Mechanical + Thermal + Chemical + Phase. Quasi-equilibrium = infinitely slow, can be drawn as a line. Cycle: $\oint d(\text{property}) = 0$. Steady: nothing inside changes with time.

Energy, heat, work (A2)

Stored E = KE + PE + U (sensible, latent, chemical, nuclear). In transit: Q (because of ΔT) and W (anything else), plus energy with mass flow.

Q, W are path functions (δQ, area under p-V), boundary phenomena. Heat or work depends on where you draw the boundary (oven example).

Pure substance (A3)

Fixed uniform composition. Water + steam = pure; liquid air + gas air = not pure (O_2 condenses first).

Phase rule $F = C - \pi + 2$. State postulate: 2 independent intensive properties.

$x = (v - v_f)/(v_g - v_f)$; $v = v_f + x\, v_{fg}$ (same for u, h, s).

Rigid tank = constant v: vertical line on T-v; $Q = m(u_2 - u_1)$.

Gases (A4)

$Pv = RT$, $R = 8.314/M$. $Z = Pv/RT$; corresponding states: $Z = f(P_R, T_R)$. Ideal if $P \ll P_{cr}$ or $T > 2T_{cr}$.

vdW $(P + a/v^2)(v - b) = RT$, $a = 27R^2T_{cr}^2/64P_{cr}$, $b = RT_{cr}/8P_{cr}$. RK: 0.42748, 0.08664, √T. N_2 175 K: ideal 13.85, vdW 9.47, RK 10.09 MPa (true 10).

First law (A5)

0th law → T; 1st → U; 2nd → S. Cycle: $\oint \delta Q = \oint \delta W$. Closed: $Q - W = \Delta U + \Delta KE + \Delta PE$.

CV: $dE/dt = \dot{Q} - \dot{W} + \Sigma \dot{m}(h + V^2/2 + gz)_{in} - \Sigma \dot{m}(\ldots)_{out}$. SFEE: $q - w = \Delta h + \Delta V^2/2 + g\Delta z$. Flow work $pv \Rightarrow h = u + pv$.

W_b: $pV^n \rightarrow (p_2V_2 - p_1V_1)/(1 - n)$; $n = 1 \rightarrow p_1V_1 \ln(V_2/V_1)$; p const → $p\Delta V$.

Devices (A6)

Turbine $w = h_1 - h_2$; compressor/pump $w_{in} = h_2 - h_1$; pump $w = v\Delta p$; heat exchanger $\dot{m}_h\Delta h_h = \dot{m}_c\Delta h_c$; throttle $h_1 = h_2$; $\dot{m} = AV/v$. KE in kJ/kg = $V^2/2000$.

Key answers: compressor 119.5 kW; pump 25.8 kW, steam 0.0616 kg/s; 2-stage turbine 17.36 kg/s, 17.6 MW, 3.82 MW; turbine 2023: 872.5 kJ/kg, 5.73 kg/s.

Second law (A7)

KP: no cycle produces work from a single reservoir. Clausius: no cycle moves heat cold → hot with no other effect. Equivalent (KP violator + fridge = Clausius violator).

Carnot principles: $\eta_{irr} < \eta_{rev}$; all reversible engines equal. $(Q_L/Q_H)_{rev} = T_L/T_H$; $\eta_C = 1 - T_L/T_H$. Proof pattern: assume opposite → reverse the reversible one → one reservoir gives work.

Entropy (A8)

Clausius inequality $\oint \delta Q/T \le 0 \Rightarrow dS = (\delta Q/T)_{rev}$. $S_2 - S_1 = \int \delta Q/T + S_{gen}$; $\Delta S_{isolated} \ge 0$.

$T\,ds = du + p\,dv$; $T\,ds = dh - v\,dp$. Ideal gas: $\Delta s = c_v \ln(T_2/T_1) + R\ln(v_2/v_1) = c_p \ln(v_2/v_1) + c_v \ln(p_2/p_1)$.

Solid/liquid: $c \ln(T_2/T_1)$.

CV: $dS/dt = \Sigma \dot{Q}/T + \Sigma \dot{m}s_{in} - \Sigma \dot{m}s_{out} + \dot{\sigma}$. $\eta_C = (h_{2s} - h_1)/(h_{2a} - h_1)$; $\eta_T = (h_1 - h_{2a})/(h_1 - h_{2s})$.

Exergy (A9) and relations (A10)

$X = (U - U_0) + p_0(V - V_0) - T_0(S - S_0) + KE + PE$.

Flow $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz$. $X_{dest} = T_0S_{gen} \ge 0$ (cannot create exergy). $W_u = W - p_0\Delta V$; $\eta_{II} = W_u/W_{rev}$.

Maxwell: $(\partial s/\partial v)_T = (\partial p/\partial T)_v$, $(\partial s/\partial p)_T = -(\partial v/\partial T)_p$, $(\partial T/\partial v)_s = -(\partial p/\partial s)_v$, $(\partial T/\partial p)_s = (\partial v/\partial s)_p$. $\beta = (1/v)(\partial v/\partial T)_p$, $\kappa = -(1/v)(\partial v/\partial p)_T$. Throttling cools only below max inversion T ($\mu_{JT} > 0$).

Cycles basics (B1)

Air-standard: air ideal gas in closed loop, reversible, combustion → heat in, exhaust → heat out; cold-air: c_p 1.005, c_v 0.718, γ 1.4.

Carnot vapour cycle impractical: T_H < 374 °C, wet turbine, two-phase compression. $\eta = 1 - T_0/T_m$, $T_m = (h_3 - h_2)/(s_3 - s_2)$.

Rankine (B2, B3)

1-2 pump, 2-3 boiler, 3-4 turbine, 4-1 condenser. $\eta = [(h_3 - h_4) - v_1\Delta p]/(h_3 - h_2)$. Improve: lower p_{cond}, superheat, higher p_{boiler}, reheat ($p_{RH} \approx$ ¼-⅕ p_B), regeneration.

Open FWH $y = (h_3 - h_2)/(h_6 - h_2)$; $\eta = 1 - (1 - y)(h_7 - h_1)/(h_5 - h_4)$.

Answers: 4 MPa/400 °C/27 °C: 38.2 % (actual 32.4 %); 3 MPa/355 °C/11 kPa: 33.2 %; reheat 15 MPa: p_{RH} 4 MPa, 45.0 %; FWH 600 °C: y 0.227, 46.3 %; 550 °C: y 0.234, 45.5 %.

IC engines (B4)

Otto $\eta = 1 - 1/r^{\gamma-1}$. Diesel $\eta = 1 - (1/r^{\gamma-1})(r_c{}^{\gamma} - 1)/[\gamma(r_c - 1)]$. Dual: $(r_p r_c{}^{\gamma} - 1)/[(r_p - 1) + \gamma r_p(r_c - 1)]$.

Same r: Otto > Dual > Diesel; same p_{max}: reverse.

Diesel uses higher r (air only, no knock). MEP = $w_{net}/(v_1 - v_2)$.

Answers: Otto r 4.89, 47 %; Diesel r 17: r_c 1.95, 867 kJ/kg, 62.6 %; r 16: 2.01, 884, 61.3 %, MEP 699 kPa; Dual 2019 57.3 %; 2018 62.7 %.

Brayton (B5) and refrigeration (B6)

$\eta = 1 - 1/r_p{}^{0.2857}$; $r_{bw} = w_c/w_t$. r_p 8: 543 K, 718 K, 44.8 %; with 80/85 %: 604 K, 805 K, 0.615, 27.4 %; r_p 5: 36.9 %, 176, 444.5 kJ/kg.

VCR: compressor → condenser → valve (h const) → evaporator; $COP = (h_1 - h_4)/(h_2 - h_1)$. VAR: absorber + pump + generator replace compressor; heat-operated, quiet, low COP. Ice plant: 48.5 kW, motor 21.5 kW.

Gas mixtures (B7)

$p_i = y_i p$ (Dalton), $V_i = y_i V$ (Amagat). $M_m = \Sigma y_i M_i = m/n$; $R_m = 8.314/M_m$; $c_p = \Sigma m f_i c_{p,i}$, $c_{p,i} = \gamma R/(\gamma - 1)$.

3 kg N_2 + 5 kg CO_2: y 0.485/0.515, M 36.24, R 0.2294, V 1.794 m^3, ρ 4.46, c_p 0.915, c_v 0.686. 3.5 + 5.5 kg: y 0.5, M 36, V 3.28, c_p 0.918.

Psychrometry (B8)

$\omega = 0.622 p_v/(p - p_v)$; $\varphi = p_v/p_s$; $\mu = \omega/\omega_s$; $\varphi = \mu/[1 - (1 - \mu)p_s/p]$; DPT = $T_{sat}(p_v)$; $h = 1.005t + \omega(2500 + 1.88t)$.

Carrier: $p_v = p_{wb} - (p - p_{wb})(t_d - t_w)/(1544 - 1.44t_w)$. Adiabatic saturation: h const. DBT > WBT (evaporation cools the wick). Humid summer ⇒ slow drying, sweaty.

35/31: 0.0424 bar, 0.0272, 75.3 %, 30 °C, 104.9 kJ/kg. 30/27: 0.0337, 0.0214, 79.4 %, 26 °C.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
BDC	bottom dead centre
CI	compression-ignition
CM	control mass
COP	coefficient of performance
CV	control volume
DBT	dry-bulb temperature
DPT	dew-point temperature
EOS	equation of state
FWH	feedwater heater
HP	high-pressure
IAPWS	International Association for the Properties of Water and Steam
IC	internal combustion
KE	kinetic energy
kmol	kilomole (molar mass in kilograms)
KP	Kelvin-Planck (statement)
kPa	kilopascal
kW	kilowatt
LP	low-pressure
MEP	mean effective pressure
MPa	megapascal
MW	megawatt
PE	potential energy
PMM1	perpetual-motion machine of the first kind
PMM2	perpetual-motion machine of the second kind
RH	relative humidity
RK	Redlich-Kwong
SFEE	steady-flow energy equation
SI	spark-ignition
SRK	Soave-Redlich-Kwong
TDC	top dead centre
Tk	Bangladeshi taka
TR	ton of refrigeration (3.5 kW)
VAR	vapour absorption refrigeration
VCR	vapour compression refrigeration
vdW	van der Waals
WBT	wet-bulb temperature